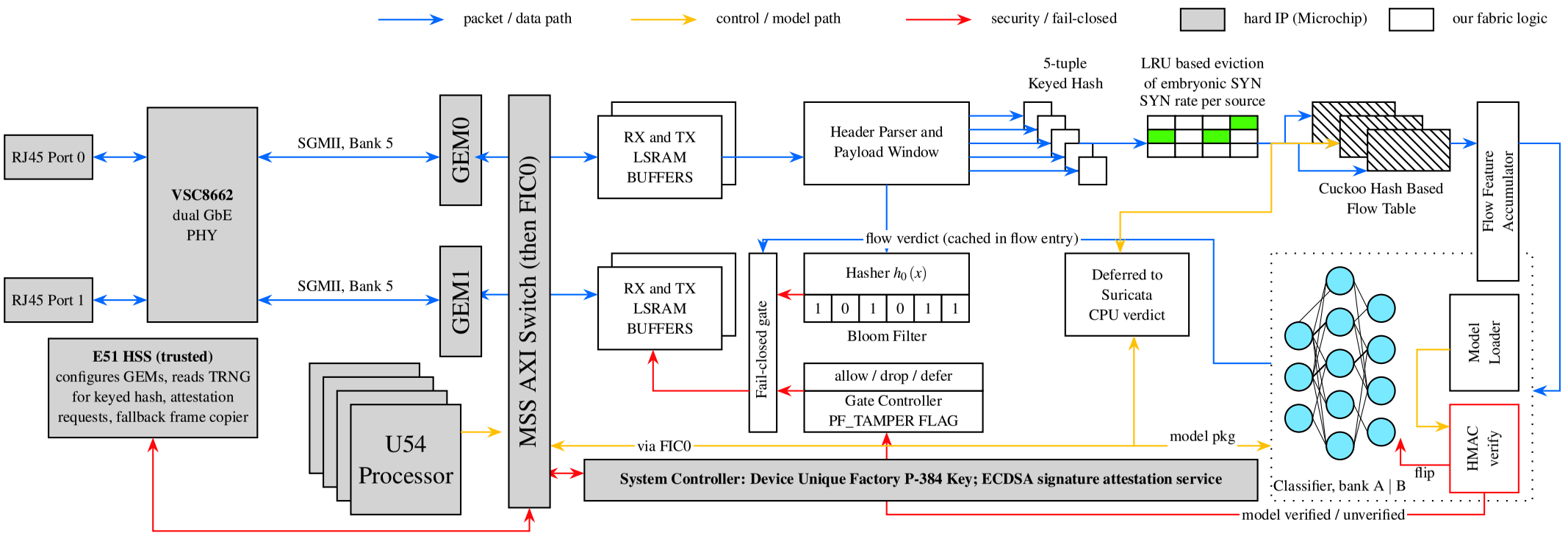


Figure 1: System block diagram: inline, fail-closed IPS on the PolarFire SoC Icicle Kit (MPFS250T). Packets enter and leave only through the VSC8662 PHY and the GEM MACs, whose buffers live in fabric LSRAM via FIC0; the fail-closed gate is the only path to TX. Linux on the U54 cores is untrusted: it can only stage authenticated model packages and return verdicts for deferred flows. The MSS AXI switch routes both GEM DMA masters to FIC0, so packet buffers can live directly in fabric LSRAM; an E51-managed frame copy is kept as a fallback.